

Lifecycle Management in Azure Local



Agenda

- **Objective:** Review Azure Local lifecycle, focusing on Azure Local Lifecycle Manager (LCM)
- **Scope:** Feature/cumulative releases, upgrade paths, validation, and hardware extension updates.



Azure Local LCM Basics

- **Arc-enabled shift (since 2311.2):**
 - Transition from a cloud-connected OS to an **Arc-enabled solution**
 - Comprising **OS, Arc,** and the **LCM**
 - Delivered as a **single solution package**.
- **IaC model & inputs:**
 - **Parameter-driven deployments** tailored to each environment
 - **LCM orchestrates** desired state across **layers**.
- **Workloads under management:**
 - Azure Arc VMs, AKS on Azure Local, Azure Virtual Desktop

LCM Role & Responsibilities

- **Central orchestrator:**
 - Drives updates, upgrades, health checks, progress tracking, and rollback/resume behaviors.
- **Compliance guardian:**
 - Helps keep clusters within the six-month support window; flags EOS risks proactively.
- **Experience unifier:**
 - Aligns Portal and PowerShell workflows; discourages out-of-band interfaces.

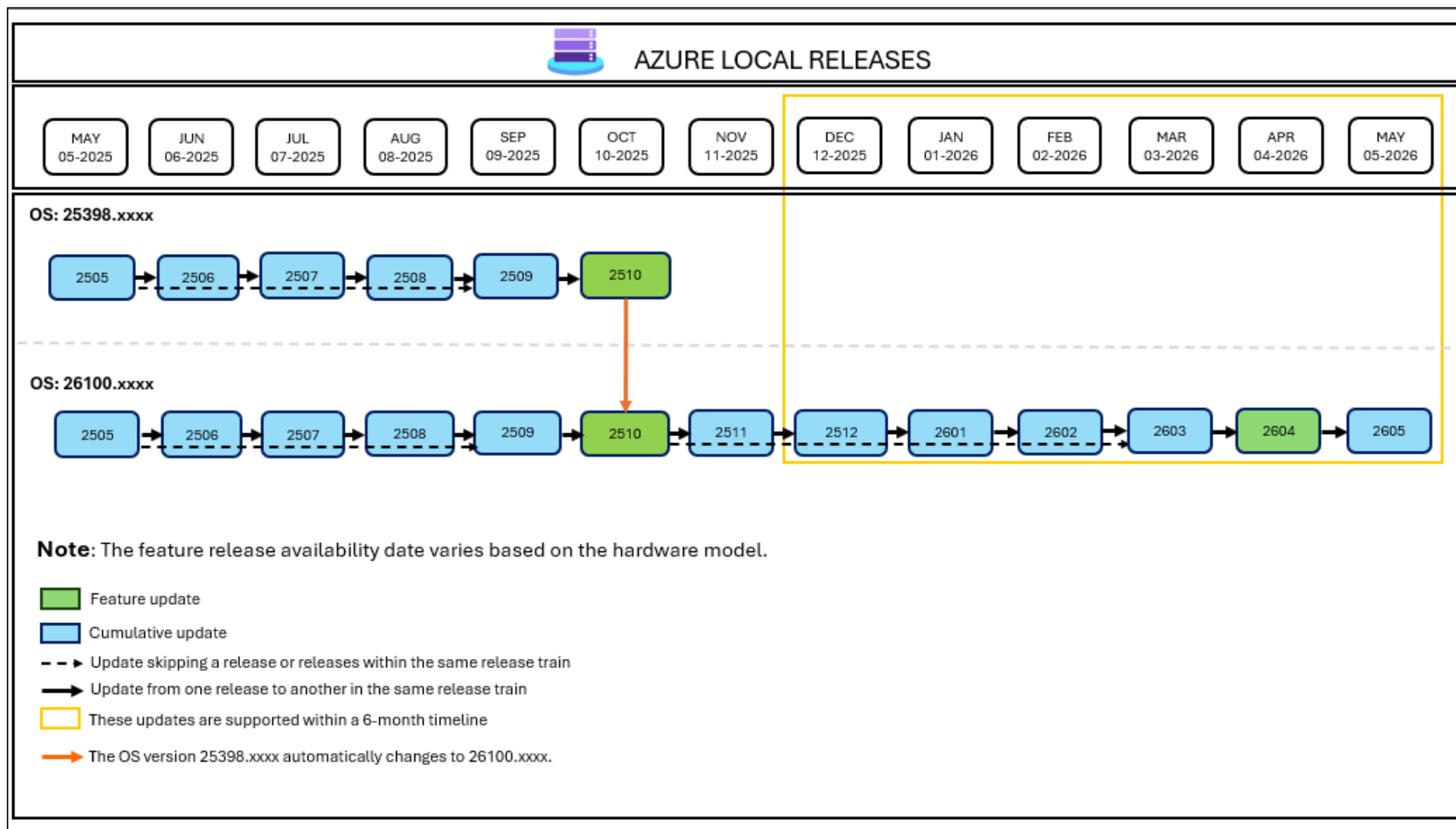
Release Cadence, Versioning & Support

- **Monthly trains (e.g., 2503, 2504, 2505...):**
 - Each train is either a **Feature build** or a **Cumulative update**.
- **Feature build:**
 - Initial version in a train; adds product enhancements, updated agents/services.
- **Cumulative update:**
 - Incremental updates after feature build; primarily **quality/security** fixes.

Versioning Changes & OS Builds

- **Post-2504 historical model:**
 - Microsoft started releasing **two solution versions per month**, each aligned to a **specific OS build**.
 - For example, 10/11.25xx aligned to 25398.xxxx and 12.25xx aligned to 26100.xxxx.
- **The 2510 fusion:**
 - The dual-servicing model ended with the 2510 release, serving as an upgrade bridge.
 - The legacy 11.xxxx track was discontinued. All subsequent updates (such as 2603, 2604, 2605) ship exclusively on a unified 12.xxxx track tied to the 26100.xxxx (24H2) OS baseline.

Azure Local Releases

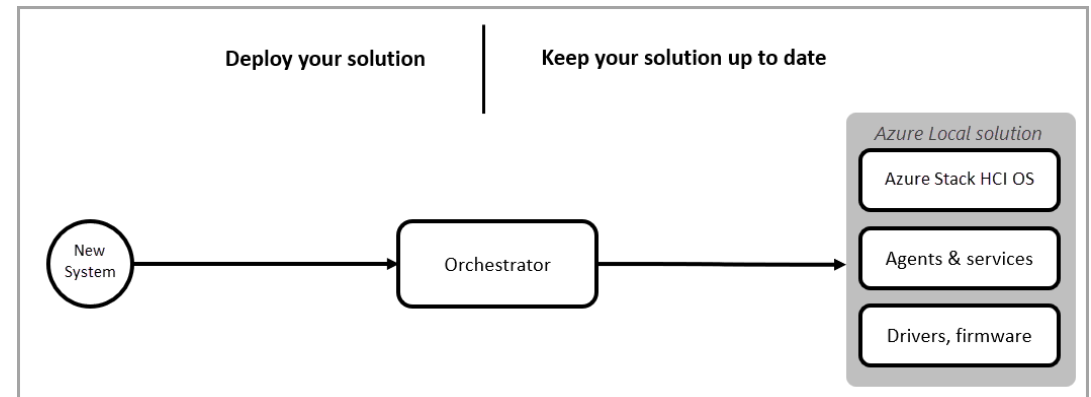
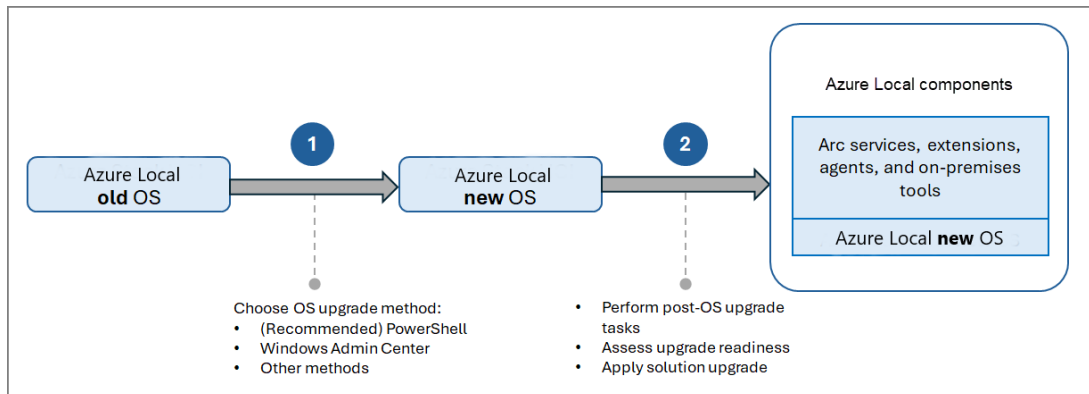


Support Windows & EOS

- **Modern Lifecycle Policy:**
 - Must stay within **six months** of the most recent release.
- **EOS impact (e.g., 23H2):**
 - Azure Stack HCI OS 23H2 reached **end of support in April 2026**; security/quality updates stop; support limited to patching to supported release.
- **Risk posture:**
 - Unsupported versions may carry **security** and **compliance** exposure;
 - Microsoft **does not guarantee remediation**.

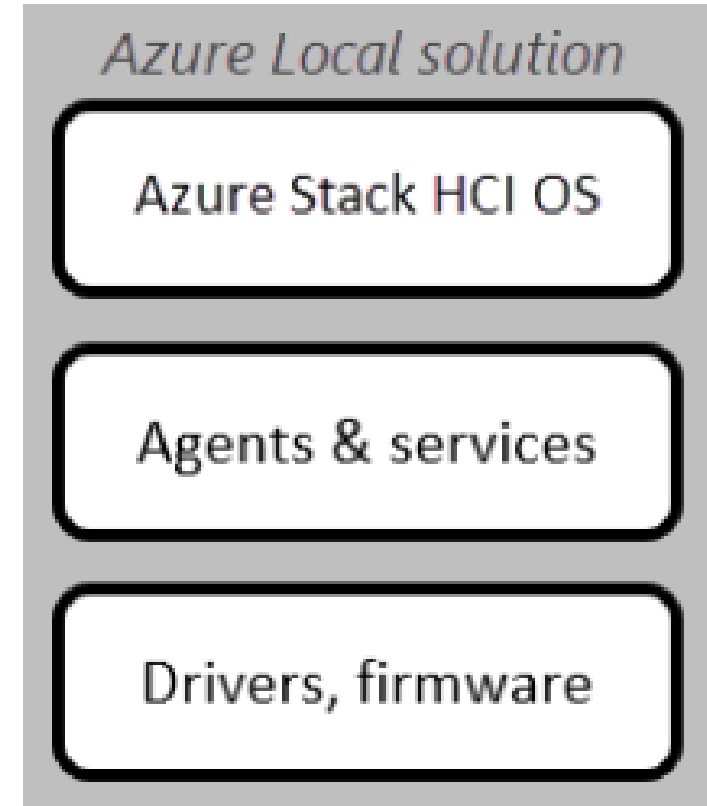
Upgrade vs. Update

- **Upgrade:**
 - Converts an older cluster (e.g., 23H2) into new **Azure Local OS**; major foundational changes.
- **Update:**
 - Moves within Azure Local (e.g., 2604 → 2605); enhances performance/security/stability.
- **Gate:**
 - Ensure **last cumulative** is installed **before** a feature update (e.g., install 2603 before 2604).



What's In a Solution Update?

- **OS:**
 - Monthly **security** and **quality** fixes
 - Mitigates **unpatched vulnerabilities**
- **Agents/services:**
 - Orchestrator
 - **Arc agent**
 - **Arc RB dependencies**
- **Solution Builder Extensions (SBE):**
 - Vendor **firmware/driver** content
 - May require **connector credentials**



SBE Hardware Updates in the Lifecycle

- **Purpose:**
 - SBE packages provide **firmware/driver** updates, **health checks**, and **download connectors** (programming interfaces allowing automatic discovery and download of future updates)
- **Integration:**
 - Runs **before/after** solution updates; **extends validation** for **deployment, update, node actions**.
- **Support scope:**
 - **Integrated Systems/Premier Solutions** implement SBE; **Validated Nodes** typically rely on **WAC**.

SBE States & Discovery

- **Key states:**
 - AppliedSuccessfully, UpdateAvailable, UpdateInProgress, NeedsAttention, PreparationFailed, AdditionalContentRequired.
- **Discovery:**
 - LCM queries vendor SBE manifests;
 - `Get-SolutionUpdate` lists ready and additional content required items.
- **Action cues:**
 - Use SBEReleaseLink, SBENotifyMessage, SupportedModels to prioritize and confirm applicability.

SBE Download Connectors & Credentials

- Automating vendor content acquisition
- Download connector:
 - Some SBE families can auto-download driver/firmware packages, reducing manual imports.
- Credentials management:
 - Provide SBE credentials during deployment
 - Alternatively, use **Set-SolutionExtensionSecretLocation** (≥ 2411).
- Benefit:
 - Updates show **Ready** rather than **AdditionalContentRequired**; easier at-scale operations.

Update Execution & Monitoring

- **Supported tools:**
 - PowerShell (LCM cmdlets), Azure Update Manager from **Azure Local** resource view (the Azure portal)
- **Unsupported:**
 - SConfig, Windows Admin Center (for solution updates), Machines (Arc) pane in the Azure portal, manual CAU runs
- **Reason:**
 - Out-of-band updates can break **lifecycle compliance**, create **unsupported states**, and cause **billing side-effects**.

Update Phases: What Happens Under the Hood

- **Preparation:**
 - Download packages, validate/extract, run health checks;
 - Update enters the **ReadyToInstall** state if healthy.
- **Installation:**
 - Orchestrated action plan across nodes;
 - UpdateRun records **steps, timings, errors**.
- **Overrides:**
 - **Start-SolutionUpdate -PrepareOnly** for pre-staging
 - **-IgnoreWarnings** to bypass warnings (use sparingly).

Monitoring Tools & Signals

- **Portal telemetry:**
 - Azure Update Manager → History/Progress;
 - Expect up to 15 minutes before runs appear.
- **PowerShell telemetry:**
 - `Get-SolutionUpdateRun` → Progress tree (steps, errors, expected durations).
- **Recovery & resume:**
 - `Start-SolutionUpdate` on failed runs;
 - For Warning states, optionally `-IgnoreWarnings`.

Troubleshooting and Resolving Workflow Issues

- **AdditionalContentRequired:**
 - Import SBE XML/ZIP bundles to Infrastructure_1\import;
 - Run **Add-SolutionUpdate**.
- **HealthCheckFailed (Critical):**
 - Address BitLocker, Network ATC, Required features;
 - Re-validate.
- **DownloadFailed:**
 - Confirm outbound connectivity, SBE connector health, Key Vault access;
 - Re-attempt.

Installation Failures & Fixes

- **Node reboot loops:**
 - Re-check BitLocker suspension
 - Verify driver versions
 - Inspect ClusterAwareUpdating logs.
- **Storage offline:**
 - Confirm ReFS key persistence
 - CSV resources started
 - pool health = Healthy.
- **Resume vs. redeploy:**
 - Prefer resume
 - Redeploy after validating UpdateRun errors are cleared.

AKS Arc Support Tool

- **Microsoft-provided PowerShell module for diagnostics and remediation:**
 - Targets infrastructure components
 - Focuses on Azure Local VMs and AKS enabled by Azure Arc on Azure Local
- **Benefits:**
 - Remediation for issues pertaining to Azure Local VMs and AKS Arc
 - Fixes for solution update problems
 - Automatically updates infrastructure components such as Microsoft On-Premises Cloud (MOC) to the latest patch release
- **Usage recommendations:**
 - Run before opening a support request, in case of the following problems:
 - Azure Local VMs runs into problems.
 - Solution update fails in MOC binaries state or in Arc resource bridge stage.
 - MOC service doesn't stay online.
 - Arc resource bridge is offline.

Best Practices: Before You Click Install

- **Review known issues:**
 - Consult **release notes** and **Known Issues** pages
 - Apply **mitigations** preemptively.
- **Test cluster strategy:**
 - Validate in **non-prod** clusters mirroring production **hardware/models**.
- **Batching & tags:**
 - Update in **small batches** first
 - Group by **location, version, subscription, resource group**.

Best Practices: Speed & Scale

- **Pre-download:**
 - Use `Start-SolutionUpdate -PrepareOnly` to cache content
 - Run readiness ahead of maintenance.
- **Limited connectivity:**
 - Download `CombinedSolutionBundle` externally
 - Import via `Add-SolutionUpdate`.
- **Large clusters (>16 nodes):**
 - Consider separating SBE updates from solution update to fit maintenance windows.

PowerShell Flow: Discover & Prepare

- Discover updates:
 - `Get-SolutionUpdate` ; Filter for Ready/AdditionalContentRequired
- Import content:
 - `Add-SolutionUpdate -SourceFolder <import>` for SBE/solution packages
- Pre-stage:
 - `Start-SolutionUpdate -PrepareOnly` ; Check State, UpdateStateProperties, HealthState

```
Get-SolutionUpdate |  
Where-Object { $_.State -like "Ready*" -or $_.State -like "Additional*" } |  
Format-List DisplayName, ResourceId, State, PackageType  
Add-SolutionUpdate -SourceFolder  
"C:\ClusterStorage\Infrastructure_1\Shares\SU1_Infrastructure_1\import"  
Get-SolutionUpdate -Id <ResourceId> | Start-SolutionUpdate -PrepareOnly
```

PowerShell Flow: Install & Monitor

- **Install:**

- `Start-SolutionUpdate` (no `PrepareOnly`) to run preparation + installation in one flow.

- **Monitor:**

- `Get-SolutionUpdateRun` to view steps, status, errors; Parse action plans for ≥ 2503

- **Resume:**

- On `InstallationFailed`, re-run `Start-SolutionUpdate` (use `-IgnoreWarnings` judiciously).

```
Get-SolutionUpdate -Id <ResourceId> | Start-SolutionUpdate  
Get-SolutionUpdate -Id <ResourceId> | Get-SolutionUpdateRun
```

Azure Update Manager Flow: Azure Portal

- **Where to go:**
 - Azure Update Manager → Resources → Azure Local
 - Select systems with Updates available and Connected.
- **Check readiness:**
 - Review readiness checks
 - Drill into Details for per-system findings & remediation guidance
- **Select updates & install:**
 - View Version details
 - Install
 - Track progress via History/Progress

Policy, Compliance & Risk

- **Policy baseline:**
 - Stay within **six months** of latest feature train (supportability)
 - Keep **Arc RB** updated within **one year** (cert validity expiration blocks updates, requiring full redeployment)
- **Portal banner cues:**
 - End of Support banners
- **Risk management:**
 - Treat **unsupported states** as **high-risk exceptions**
 - Prioritize returning to **supported**.

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